



Week 4 Workshop

IR models and $tf*idf$

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Workshop Objectives

1. **The first objective** is to understand how the TF*IDF technique is applied to both documents and queries, including the definition and computation of document frequency (df).
2. **The second objective** is to understand contingency tables and the associated variables (e.g., n_i and r_i) used in the Binary Independence Model to represent the distribution of terms in relevant and non-relevant documents.
 - These concepts form the foundation of the BM25 ranking model.

Task 1

TF*IDF is a term-weighting scheme that combines **term frequency (tf)** and **inverse document frequency (idf)** to quantify the importance of a term within a document relative to a document collection. While the underlying intuition of TF*IDF is consistent, multiple formulations exist that differ in how term frequency, inverse document frequency, and document length normalization are defined.

weighting scheme	document term weight	query term weight
1	$f_{t,d} \cdot \log \frac{N}{n_t}$	$\left(0.5 + 0.5 \frac{f_{t,q}}{\max_t f_{t,q}}\right) \cdot \log \frac{N}{n_t}$
2	$1 + \log f_{t,d}$	$\log \left(1 + \frac{N}{n_t}\right)$
3	$(1 + \log f_{t,d}) \cdot \log \frac{N}{n_t}$	$(1 + \log f_{t,q}) \cdot \log \frac{N}{n_t}$

Discuss the TF*IDF weighting scheme recommended below and the scheme presented in the lecture notes. Your discussion should compare the two approaches with respect to:

- how **term frequency** is computed (e.g., raw counts, normalized frequency, or logarithmic scaling),
- how **inverse document frequency** is defined (e.g., smoothed vs. unsmoothed formulations),
- and how **document length normalization** is handled (e.g., cosine normalization, length-normalized TF, or no normalization).

*Recap: the heuristic modification of tf*idf weight in week 3 lecture notes*

$$d_{ik} = \frac{(\log(f_{ik}) + 1) \cdot \log(N/n_k)}{\sqrt{\sum_{x=1}^t [(\log(f_{ix}) + 1) \cdot \log(N/n_x)]^2}}$$

Task 2

Manually calculate the document frequency df value for each term in the following table.

Document	term1	term2	term3	term4	term5
D1	3	0	0	5	7
D2	5	3	4	6	0
D3	0	0	5	4	6
D4	9	0	0	1	2
D5	0	1	0	3	2
D6	3	0	2	4	4
df					

Recap: the Python function to calculate df for each term in week 3 lecture notes, where *docs* is a Python dictionary in which each element (e.g., a document) is represented as a set of *term:frequency* pairs.

```
def c_df(docs):  
    df_ = {}  
    for id, doc in docs.items():  
        for term in doc.keys():  
            try:  
                df_[term] += 1  
            except KeyError:  
                df_[term] = 1  
    return df_
```

Task 3

Design a python function `c_df(docs)` to calculate *df* value for each term in *docs* to verify if you can get the same result as you did in Task 2. The function returns a {term:df, ...} dictionary. In your program, you can represent the above table as follows when you use it to test your python function.

```
docs = {'D1':{'term1':3, 'term4':5, 'term5':7}, 'D2':{'term1':5, 'term2':3,
'term3':4, 'term4':6}, 'D3':{'term3':5, 'term4':4, 'term5':6}, 'D4':{'term1':9,
'term4':1, 'term5':2}, 'D5':{'term2':1, 'term4':3, 'term5':2}, 'D6':{'term1':3,
'term3':2, 'term4':4, 'term5':4}}
```

Hint: see the python function defined in week 3 lecture notes or recapped in task 2

Task 4

Let $Q = \{\text{US}, \text{ECONOM}, \text{ESPIONAG}\}$ be a query, and

$\mathbf{C} = \{D_1, D_2, D_3, D_4, D_5, D_6, D_7\}$ be a collection of documents, where

$D_1 = \{\text{GERMAN}, \text{VW}\}$

$D_2 = \{\text{US}, \text{US}, \text{ECONOM}, \text{SPY}\}$

$D_3 = \{\text{US}, \text{BILL}, \text{ECONOM}, \text{ESPIONAG}\}$

$D_4 = \{\text{US}, \text{ECONOM}, \text{ESPIONAG}, \text{BILL}\}$

$D_5 = \{\text{GERMAN}, \text{MAN}, \text{VW}, \text{ESPIONAG}\}$

$D_6 = \{\text{GERMAN}, \text{GERMAN}, \text{MAN}, \text{VW}, \text{SPY}\}$

$D_7 = \{\text{US}, \text{MAN}, \text{VW}\}$

Assume relevant and non-relevant documents (user feedback) are labeled as follows:

Document ID	Terms: d_{ij}	Relevance to Q
D_1	GERMAN, VW	0 no
D_2	US, US, ECONOM, SPY	1 yes
D_3	US, BILL, ECONOM, ESPIONAG	1 yes
D_4	US, ECONOM, ESPIONAG, BILL	1 yes
D_5	GERMAN, MAN, VW, ESPIONAG	0 no
D_6	GERMAN, GERMAN, MAN, VW, SPY	0 no
D_7	US, MAN, VW	0 no

Task 4 cont.

For a given incoming document $D = \{\text{US}, \text{VW}, \text{ESPIONAG}\}$, let *term 1* = 'US', *term 2* = 'VW' and *term 3* = 'ESPIONAG'. Based on binary independence model, work out the missing values for the following contingency tables, where $d_i = 1$ if term i is present in the document, and 0 otherwise.

	Relevant	Non-relevant	Total
$d_1 = 1$	$r_i = 3$	$n_i - r_i = 1$	$n_i = 4$
$d_1 = 0$	$R - r_i = 0$	$(N-R) - (n_i - r_i) = N - n_i - R + r_i = 3$	$N - n_i = 3$
Total	$R = 3$	$N - R = 4$	$N = 7$

	Relevant	Non-relevant	Total
$d_2 = 1$	$r_i =$	$n_i - r_i =$	$n_i =$
$d_2 = 0$	$R - r_i =$	$N - n_i - R + r_i =$	$N - n_i =$
Total	$R =$	$N - R =$	$N =$

	Relevant	Non-relevant	Total
$d_3 = 1$	$r_i =$	$n_i - r_i =$	$n_i =$
$d_3 = 0$	$R - r_i =$	$N - n_i - R + r_i =$	$N - n_i =$
Total	$R =$	$N - R =$	$N =$

Recap: contingency table in week 3 lecture notes

	Relevant	Non-relevant	Total
$d_i = 1$	r_i	$n_i - r_i$	n_i
$d_i = 0$	$R - r_i$	$N - n_i - R + r_i$	$N - n_i$
Total	R	$N - R$	N

- r_i is the number of relevant documents containing term i
- n_i is the number of documents containing term i
- N is the total number of documents in the collection
- R is the number of relevant documents for this query
- $d_i = 1$ if term i is present in the document, and 0 otherwise